

Skills Summary

Rewriting Expressions with Distributive Structure

Use this reference while completing your worksheet or when studying.

1. Distribute a factor over a sum

Rule: $a(b + c) = ab + ac$ and $a(b - c) = ab - ac$.

Model: a rectangle a tall and $(b + c)$ wide splits into an a -by- b piece and an a -by- c piece. One rectangle, two pieces, same area — that is the whole rule.

The multiplier reaches *every* term inside the parentheses, and each term keeps its own sign.

Example: Expand $7(3x + 4)$.

Step 1. One factor outside, a sum inside — distribute, so the 7 multiplies both terms.

Step 2. $7 \cdot 3x = 21x$ and $7 \cdot 4 = 28$, then write them as a sum.

$$\Rightarrow 7(3x + 4) = 21x + 28$$

Try it: Expand $5(2x - 9)$.

check: $10x - 45$

2. Expand a product of two binomials

Rule: $(x + p)(x + q) = x^2 + qx + px + pq$ — distribute twice, so every term of the first factor meets every term of the second (four products, then combine the two like terms).

Model: a rectangle $(x + p)$ by $(x + q)$ cuts into four smaller rectangles: the x -by- x square, two strips, and the p -by- q corner.

Example: Expand $(x + 4)(x + 6)$.

Step 1. Both factors are sums, so the same rule is applied twice — there is no single number outside to distribute.

Step 2. The four products are $x \cdot x = x^2$, $6x$, $4x$ and 24 ; the middle two are like terms, and $6x + 4x = 10x$.

$$\Rightarrow (x + 4)(x + 6) = x^2 + 10x + 24$$

Try it: Expand $(x + 2)(x + 7)$.

check: $x^2 + 9x + 14$

3. Factor out the greatest common factor

Rule: $ab + ac = a(b + c)$ — factoring is distribution read right-to-left.

How to find the GCF: take the largest number that divides every coefficient, times the highest power of the variable that appears in *every* term. Then divide each term by it and write the quotients inside the parentheses. Distribute back to check.

Example: Factor $8x + 20$.

Step 1. Both terms are being added and share a numerical factor, so pull out the greatest common factor rather than expanding anything.

Step 2. The GCF of 8 and 20 is 4; there is no x in the second term, so no x comes out. Divide: $8x \div 4 = 2x$ and $20 \div 4 = 5$.

$$\Rightarrow 8x + 20 = 4(2x + 5)$$

Try it: Factor $9x^2 + 15x$ completely.

$$(3x + 5)3x$$

4. Check whether two forms are equivalent

Two ways to decide. (1) **Rewrite:** expand and combine like terms until both sides are in the form $ax + b$; equal forms means equivalent expressions. (2) **Test a value:** substitute the same number for x in both. Equivalent expressions agree at every value, so one disagreement proves they are different — but agreement at one value is not a proof.

Example: Write $3(2x + 5) + 4x$ in the form $ax + b$.

Step 1. Parentheses first, loose terms after — distribute, then collect like terms.

Step 2. $3(2x + 5) = 6x + 15$; adding $4x$ gives $6x + 4x = 10x$ and the constant 15 is unchanged.

$$\Rightarrow 3(2x + 5) + 4x = 10x + 15$$

Try it: Write $2(4x - 3) + 5$ in the form $ax + b$.

$$8x - 1$$

(!) **Watch out:** distributing to the first term only is the most common error — $5(2x - 3)$ is $10x - 15$, never $10x - 3$. And “factored” means *completely* factored: $5(3x^2 + 2x)$ still hides a shared x .